

Report: Twitter Financial News Classification

1 Introduction

This report summarizes the development and evaluation of neural network models for classifying financial news tweets into one of 20 predefined categories (e.g., *Analyst Update*, *Earnings*, *Stock Movement*). The dataset consists of 16,990 training samples and 4,117 validation samples, each comprising a tweet text and a label (0–19). The primary challenge is severe class imbalance, which was addressed using inverse class weighting during training.

2 Data Preprocessing

Raw tweets were cleaned by removing hyperlinks and punctuation, followed by lemmatization using NLTK’s WordNetLemmatizer. Two feature representations were prepared:

- **Bag-of-Words (CountVectorizer)** with a vocabulary of 18,000 tokens, used for a feed-forward neural network (ANN).
- **Tokenized sequences** with a vocabulary of 30,000 words, padded to a maximum length of 64 tokens, used for recurrent and convolutional models.

A word cloud of the most frequent terms is shown in Figure 1, and the imbalanced label distribution is illustrated in Figure 2.

3 Model Architectures

Three neural network architectures were implemented and trained for 100 epochs with a batch size of 32. All models used the Adam or SGD optimizer with categorical cross-entropy loss and inverse class weights derived from the training set.

3.1 Simple Recurrent Neural Network (RNN)

An embedding layer (dimension 64) followed by a SimpleRNN layer with 12 units (returning sequences), global max pooling, and two dense layers (64 and 20) with dropout (0.5) was employed.

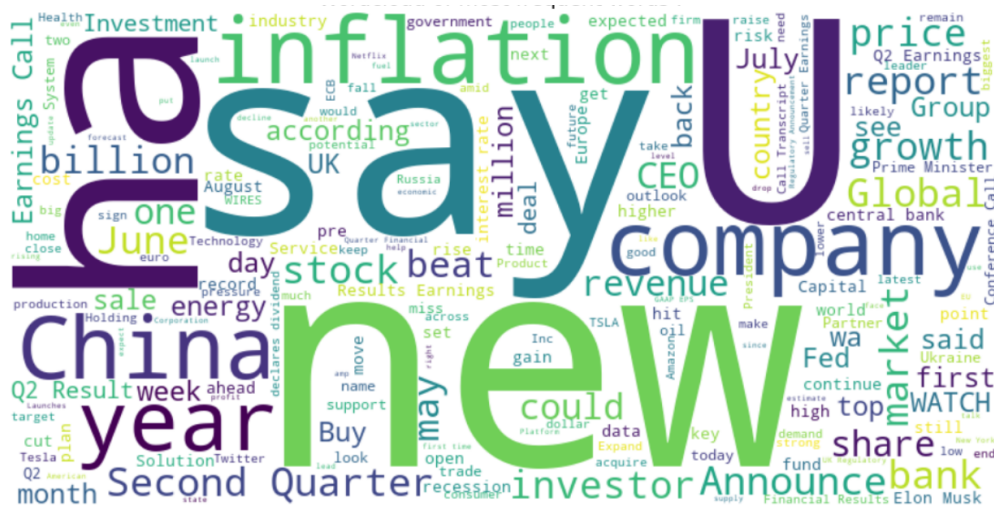


Figure 1: Word cloud of the training corpus after cleaning.

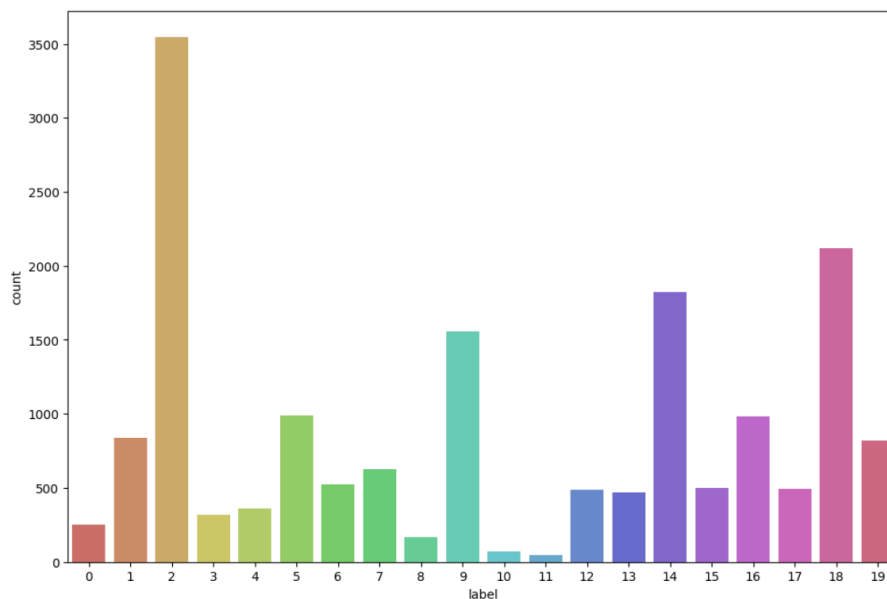


Figure 2: Distribution of training labels (highly imbalanced).

3.2 Convolutional Neural Network (CNN)

An embedding layer (64 dimensions) was followed by a 1D convolutional layer (32 filters, kernel size 3), global max pooling, and dense layers (64 and 20) with dropout (0.3). This model was compiled with SGD (learning rate 0.0001, momentum 0.9).

3.3 Feed-Forward Neural Network (ANN)

A three-hidden-layer network (1028, 512, 256 units) with dropout (0.2) was built on the bag-of-words representation (18,000 features). It also used SGD with the same hyperparam-

eters.

4 Results

Table 1 summarizes the final validation accuracy and macro F1-score for each model. The ANN achieved the best overall performance, reaching 83% accuracy and a macro F1 of 0.81. The CNN and RNN underperformed, likely due to the limited sequence length and the high imbalance.

Table 1: Model performance on the validation set.

Model	Accuracy	Macro F1
SimpleRNN	0.78	0.78
CNN	0.78	0.78
ANN	0.83	0.81

Detailed classification reports for the ANN and CNN are provided in Appendix A. The confusion matrices indicate that while the model performs well on majority classes, performance on minority classes (e.g., class 10 and 11) remains challenging despite class weighting.

5 Conclusion

A feed-forward neural network using a bag-of-words representation yielded the highest accuracy (83%) on the Twitter financial news dataset. Sequence-based models (RNN, CNN) did not outperform the simpler ANN, suggesting that for this task, word presence is more informative than word order. Further improvements may be obtained by employing more sophisticated architectures (e.g., Transformers) or advanced data augmentation techniques to handle class imbalance.

Classification Reports

ANN (best model)

	precision	recall	f1-score	support
0	0.67	0.86	0.75	57
1	0.86	0.88	0.87	209
2	0.85	0.83	0.84	870
3	0.75	0.92	0.83	63
4	0.99	0.96	0.97	100
5	0.94	0.88	0.91	259
6	0.79	0.82	0.81	141
7	0.83	0.86	0.85	154
8	0.84	0.77	0.81	35

9	0.78	0.70	0.74	375
10	0.62	0.38	0.47	21
11	0.93	0.81	0.87	16
12	0.79	0.87	0.83	108
13	0.70	0.82	0.75	99
14	0.83	0.81	0.82	422
15	0.77	0.76	0.76	127
16	0.87	0.91	0.89	239
17	0.82	0.85	0.84	108
18	0.85	0.85	0.85	527
19	0.73	0.76	0.74	187
micro avg	0.83	0.83	0.83	4117
macro avg	0.81	0.82	0.81	4117
weighted avg	0.83	0.83	0.83	4117